

Assignment 6

Bully.java

```
import java.util.Scanner;

public class Bully
{
    static boolean[] state = new boolean[6];
    static int coordinator;

    public static void up(int up)
    {
        if (state[up - 1])
        {
            System.out.println("Process " + up + " is already up");
        } else {
            state[up - 1] = true;
            System.out.println("Process " + up + " held election");
            for (int i = up; i < 5; i++)
            {
                System.out.println("Election message sent from process " + up + " to process " + (i + 1));
            }
            for (int i = 4; i >= 0; i--)
            {
                if (!state[i]) continue;
                System.out.println("Alive message sent from process " + (i + 1) + " to process " + up);
                coordinator = i + 1;
                break;
            }
        }
    }

    public static void down(int down)
    {
        if (!state[down - 1])
        {
            System.out.println("Process " + down + " is already down!");
        } else {
            state[down - 1] = false;
            System.out.println("Process " + down + " is down.");
        }
    }

    public static void mess(int mess)
    {
        if (!state[mess - 1])
        {
            System.out.println("Process is down");
        }
        else
        {
            System.out.println("Process " + mess + " sent an election message");
        }
    }
}
```

```

        for (int i = mess; i < 5; i++)
        {
            System.out.println("Election message sent from process " + mess + " to process " + (i +
1));
        }
        System.out.println("Coordinator message sent from process " + coordinator + " to all");
    }
}

public static void main(String[] args)
{
    Scanner sc = new Scanner(System.in);
    for (int i = 0; i < 5; i++)
    {
        state[i] = true;
    }
    coordinator = 5;
    System.out.println("5 active processes are: p1 p2 p3 p4 p5");
    System.out.println("Process 5 is the coordinator");

    while (true)
    {
        System.out.println("\n1. Up a process\n2. Down a process\n3. Send a message\n4. Exit");
        int choice = sc.nextInt();
        switch (choice)
        {
            case 1:
                System.out.println("Enter process number to bring up:");
                int up = sc.nextInt();
                up(up);
                break;
            case 2:
                System.out.println("Enter process number to bring down:");
                int down = sc.nextInt();
                down(down);
                break;
            case 3:
                System.out.println("Enter process number to send a message:");
                int mess = sc.nextInt();
                mess(mess);
                break;
            case 4:
                System.out.println("Exiting...");
                sc.close();
                return;
            default:
                System.out.println("Invalid choice, try again.");
        }
    }
}
}

```

Output:

5 active processes are: p1 p2 p3 p4 p5

Process 5 is the coordinator

1. Up a process
2. Down a process
3. Send a message
4. Exit

Enter choice:

2

Enter process number to bring down:

2

Process 2 is down.

1. Up a process
2. Down a process
3. Send a message
4. Exit

Enter choice:

3

Enter process number to send a message:

3

Process 3 sent an election message

Election message sent from process 3 to process 4

Election message sent from process 3 to process 5

Coordinator message sent from process 5 to all

1. Up a process
2. Down a process
3. Send a message
4. Exit

Enter choice:

1

Enter process number to bring up:

2

Process 2 held election

Election message sent from process 2 to process 3

Election message sent from process 2 to process 4

Election message sent from process 2 to process 5

Alive message sent from process 5 to process 2

1. Up a process
2. Down a process
3. Send a message
4. Exit

Enter choice:

4

Exiting...

Ring.java

```
import java.util.Scanner;

class Process {
    int id;
    boolean active;

    Process(int id) {
        this.id = id;
        this.active = true;
    }
}

public class Ring {
    static Process[] processes;
    static int n;

    public static void main(String[] args) {
        Scanner in = new Scanner(System.in);

        // Step 1: Initialize Processes
        System.out.print("Enter number of processes: ");
        n = in.nextInt();
        processes = new Process[n];

        System.out.println("Enter process IDs:");
        for (int i = 0; i < n; i++) {
            processes[i] = new Process(in.nextInt());
        }

        int choice;
        do {
            System.out.println("\nMENU:");
            System.out.println("1. Start Election");
            System.out.println("2. Exit");
            System.out.print("Enter your choice: ");
            choice = in.nextInt();

            switch (choice) {
                case 1:
                    System.out.print("Enter process index (0-" + (n - 1) + ") to start election: ");
                    int initiator = in.nextInt();
                    startElection(initiator);
                    break;
                case 2:
                    System.out.println("Exiting...");
                    break;
                default:
                    System.out.println("Invalid choice! Try again.");
            }
        }
```

```

    } while (choice != 2);

    in.close();
}

static void startElection(int initiator) {
    int coordinator = -1;
    System.out.println("\nElection started by Process " + processes[initiator].id);

    int i = initiator;
    do {
        int next = (i + 1) % n; // Move to the next process in the ring
        if (processes[next].active) {
            System.out.println("Process " + processes[i].id + " sends message to Process " +
processes[next].id);
            coordinator = Math.max(coordinator, processes[next].id);
        }
        i = next;
    } while (i != initiator); // Complete one full cycle

    System.out.println("\nProcess " + coordinator + " is elected as Coordinator.");
}
}

```

Output:

Enter number of processes: 5

Enter process IDs:

5

2

9

4

2

MENU:

1. Start Election

2. Exit

Enter your choice: 1

Enter process index (0-4) to start election: 2

Election started by Process 9

Process 9 sends message to Process 4

Process 4 sends message to Process 2

Process 2 sends message to Process 5

Process 5 sends message to Process 2

Process 2 sends message to Process 9

Process 9 is elected as Coordinator.

MENU:

1. Start Election

2. Exit

Enter your choice: 2

Exiting...